IOWA STATE UNIVERSITY | Ivy College of Business

Final Project Report

on

LinkedIn Job Market Insights: Trends & Patterns

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Table of Contents

Introduction 3
Dataset Details 3
Variables 4
Data Cleaning and Preprocessing 5
Exploratory Data Analysis (EDA) 5
Salary Prediction Using ML Models 6
Modeling Approach 8
Key Analytical Findings 9
Discussion 42
Limitations 42
Conclusion 43
Explore More on Our Blog: 43

Introduction

The labor market is rapidly evolving, with companies and job seekers relying heavily on online platforms like LinkedIn for recruitment. This project aims to analyze a comprehensive dataset of 124,000+ LinkedIn job postings from 2023-2024 to uncover key insights into hiring trends, salary patterns, skill demands, and industry growth. By leveraging advanced business analytics and machine learning, we can help job seekers, recruiters, and policymakers make data-driven decisions.

Dataset Details

Source: We used a publicly available dataset from Kaggle that pulls together everything from job titles and salaries to benefits, required skills, and company details. over 124,000 job postings listed on LinkedIn from 2023 to 2024.

https://www.kaggle.com/datasets/arshkon/linkedin-job-postings/data

https://iastate.box.com/s/s10fjtiojv8ozl5o2ayftfizjaa2re8c (Refer this link to download Raw and clean CSV files for smooth running of python script)

It included:

- Job postings with descriptions, salaries, and locations
- Company profiles and employee counts
- Required skills and certifications
- Benefits like stock options, PTO, and 401(k)s

Files in the Dataset:

1. **Postings.csv:** Contains core job listing information such as job title, description, location, salary, experience level, and remote status.
2. **Skills.csv:** Provides the full list of skills referenced across job postings, with abbreviations and full names.
3. **job_skills.csv:** Maps each job to its associated skills using skill abbreviations.

4. **Industries.csv:** Lists all industry categories that jobs or companies may be associated with.
5. **job_industries.csv:** Connects job postings to one or more industries using industry IDs.
6. **Benefits.csv:** Contains information about benefits offered for each job (e.g., 401(k), stock options, PTO, insurance).
7. **Salaries.csv:** Additional salary data (min, max, med) for job listings, possibly including externally enriched values.
8. **Companies.csv:** Contains company-level information such as name, location, company size, and website URL.
9. **company_industries.csv:** Links companies to the industries they operate in.
10. **company_specialities.csv:** Provides information on the areas of specialization for each company.
11. **employee_counts.csv:** Contains historical headcount data per company, helpful for analyzing hiring patterns and growth trends.

Variables

- **Job-level variables:** title, job description, remote status, salary ranges (min/max/med), listed time, work type, company ID, experience level, number of views/applies
- **Company-level variables:** company name, size, state, industry, specialties, follower count
- **Skill-level variables:** skill name, abbreviation, frequency
- **Geographic variables:** location, zip code, FIPS code, country
- **Industry and benefit variables:** industry category, benefit type (e.g., stock, insurance), mapped through foreign keys

Custom Variables:

- **estimated_salary:** Computed as the average of min_salary and max_salary where available.
- **dominant_skill_type:** Labeled jobs as "hard skill-dominant" or "soft skill-dominant" based on required skills.
- **state_extracted:** Parsed from the location column to get U.S. state abbreviations for location-based analysis.
- **quarter:** Derived from posting date to examine seasonality

Data Cleaning and Preprocessing

Data cleaning and preprocessing were critical to ensure high-quality inputs for analysis and modeling. Below are the specific steps taken:

- **Missing Values:** Dropped rows with missing or invalid salary entries, especially where min, max, and normalized salaries were all unavailable.
- **Outlier Removal:** Removed salary outliers by setting logical caps and floors (e.g., excluding salaries above $500k and below $20k).
- **Location Parsing:** Standardized location entries and extracted U.S. state abbreviations into a new state_extracted column. This enabled regional salary comparison.
- **Datetime Conversion:** Converted listed_time and closed_time to datetime format and extracted
- **Merging Datasets:** Merged multiple datasets using foreign keys such as job_id and company_id to incorporate company size, industry, benefits, and skills.
- **Skill Normalization:** Cleaned and standardized skill names by converting to lowercase, removing duplicates, and mapping abbreviations to full skill names.
- **Text Processing:** Cleaned description and title fields by removing HTML tags, special characters, and stop words. TF-IDF vectors were generated for clustering and keyword-based salary modeling.

Exploratory Data Analysis (EDA)

EDA provided the foundation for understanding overall patterns in the dataset and guiding further modeling. The key analyses performed include:

- **Salary Distribution:** Boxplots and histograms were created to explore the distribution of max, min, and normalized salaries across different job types, experience levels, and benefit types.
- **Skill Insights:** Aggregated and ranked skills by frequency and by their associated average estimated salaries. Identified top-paying skills (e.g., product management, marketing) and commonly listed ones (e.g., communication, Excel).

- **COL-Adjusted Salaries:** Created state-level plots comparing normalized salaries. Found that Midwest states like Indiana and South Carolina offered high real income after adjusting for cost of living.
- **Remote Work Patterns:** Bar plots showed percentage of remote jobs by state. Found remote work was more common in metro areas like San Francisco, Austin, and Charlotte. Remote jobs often had competitive or even higher salaries.
- **Benefit Impact:** Analyzed how benefits such as 401(k), stock options, and medical insurance affected median salary. Jobs listing benefits had noticeably higher compensation in many industries.
- **Clustering by Role Type:** Used KMeans and TF-IDF on job descriptions to cluster similar job roles. Clusters revealed distinct job families such as tech, healthcare, finance, and education.

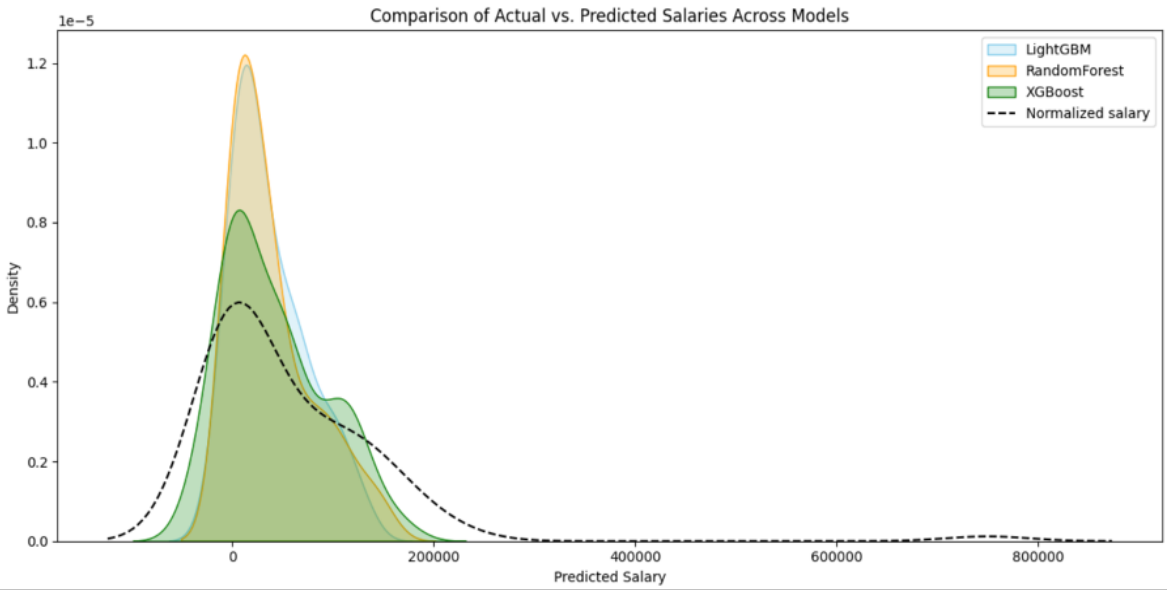
Salary Prediction Using ML Models

Objective: To predict salaries using structured job posting features by training and comparing three machine learning models: Random Forest, XGBoost, and LightGBM.

Approach: We built a regression pipeline to estimate normalized_salary using features such as job title, location, experience level, remote status, and work type. The key steps were:

- **Data Preparation**
  Cleaned the job postings data, removed missing values, and normalized salaries to a yearly basis.
- **Feature Selection & Encoding**
  Selected categorical and binary variables and encoded them using LabelEncoder.
- **Model Training**
  Split the data into training and testing sets. Trained three models with default hyperparameters:
  - RandomForestRegressor
  - XGBRegressor
  - LGBMRegressor

Model Comparison (KDE Plot): The KDE plot compares predicted salary distributions from each model with the actual salary distribution



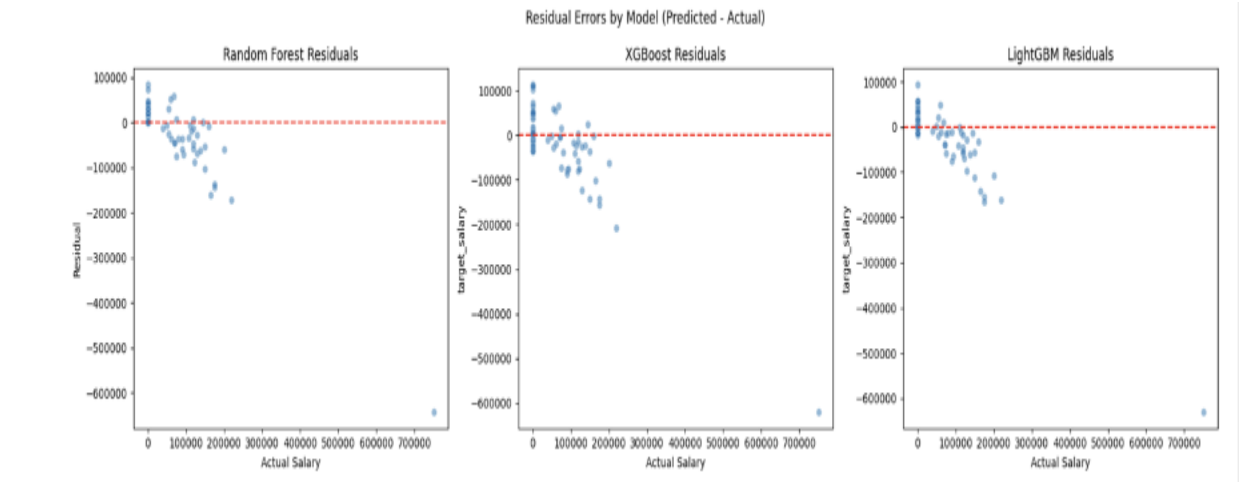
- **XGBoost** (green): Best matched real-world salary shape, slight underprediction at the high end.
- **Random Forest** (orange): Captures variation but peaks earlier than actuals.
- **LightGBM** (blue): Most conservative — clusters around the average and underpredicts top salaries.

Performance Metrics:

- **Mean Absolute Error (MAE)**: ~$47,930
- **R^2 Score**: ~0.17 — The model explains 17.2% of salary variance.

Interpretation:
All models show strong mid-range performance but tend to **underpredict high salaries**, highlighting the complexity of salary estimation from limited structured features.

Residual Plot Analysis: The residual plots reveal how prediction errors vary across actual salary ranges.



- **Random Forest**: High error spread especially underestimates high salaries.
- **XGBoost**: Balanced, slightly tighter residual spread.
- **LightGBM**: Most consistent on low/mid salaries but underperforms at the high end.

While all models capture general salary patterns, XGBoost emerges as the most balanced performer. However, the relatively low R^2 score suggests that salary is influenced by additional unobserved factors (e.g., negotiation, education, company prestige) that were not captured in the dataset.

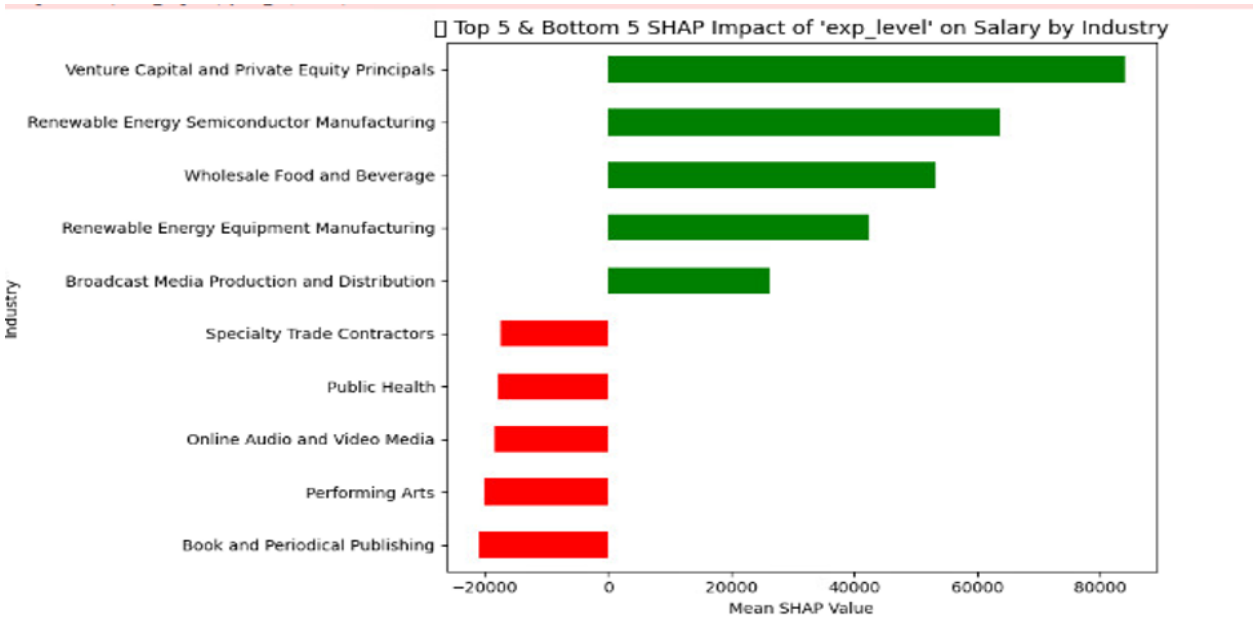
Modeling Approach

To uncover patterns in job postings, we applied a mix of machine learning and statistical techniques. Regression models were used to predict salary ranges based on features like job title, location, and skills. We used KMeans clustering on skills and job descriptions to identify hidden job groups and hybrid roles. TF-IDF was applied to extract influential keywords from job descriptions. SHAP analysis helped interpret feature importance and contribution to salary prediction. For seasonality trends, we grouped data by quarters and experience levels to detect hiring cycles.

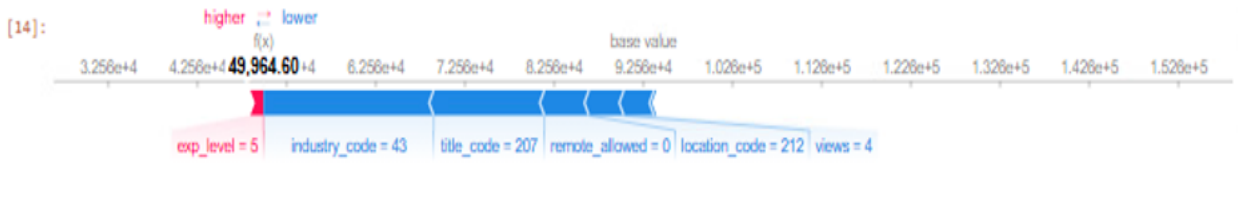
Key Analytical Findings

1. **Salary Prediction Factors:**

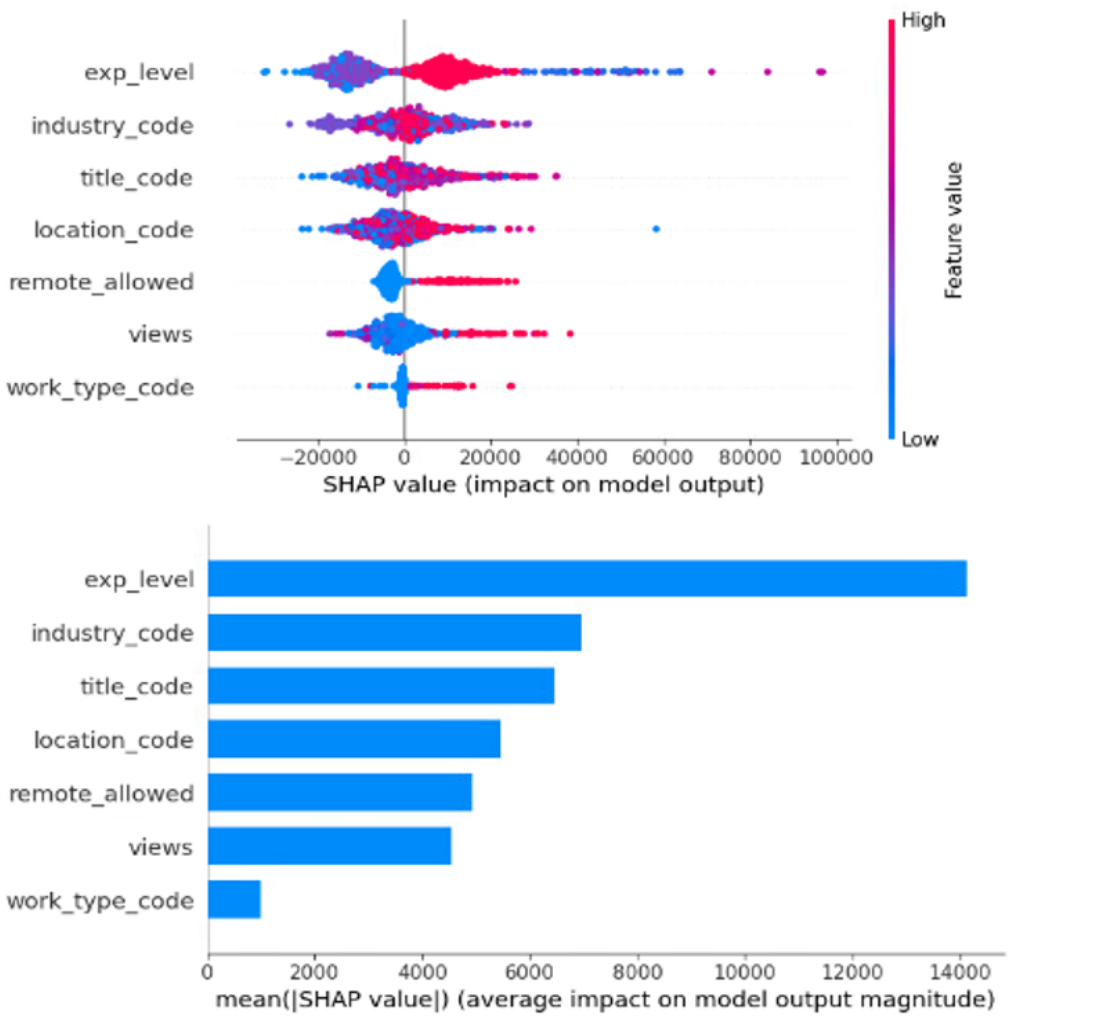
   Using machine learning models like XGBoost and SHAP explainability, we found that **experience level** is the strongest driver of salary, followed by **industry type** and **job title**. Features like **remote work**, **location**, and **work type** had varied impacts across industries.



- **Venture Capital**, **Renewable Energy**, and **Wholesale sectors** show the **highest positive SHAP values** for experience level, indicating that in these industries, **experience significantly boosts salary**.
- **Performing Arts**, **Publishing**, and **Public Health** showed **negative SHAP impacts**, meaning additional experience doesn't lead to higher salary predictions in these industries — possibly due to budget constraints or flat pay structures.
- ***Insight:*** Industry context matters — in high-growth, capital-heavy sectors, experience is rewarded more aggressively.



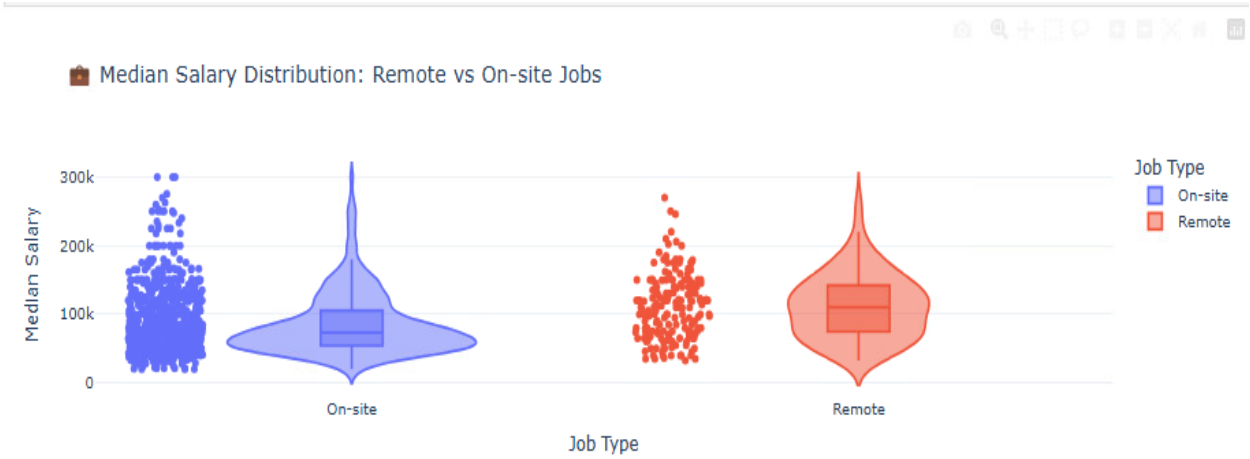
- This visual shows **how different features push an individual salary prediction up or down**.
- In the shown case, exp_level = 5 adds the **most positive contribution**, while remote_allowed = 0 (i.e., on-site job) and the job's **location** pull the salary down.
- ***Insight:*** A candidate with high experience in a non-remote, low-paying location sees their potential capped despite their experience — remote roles and better location coding could improve salary.



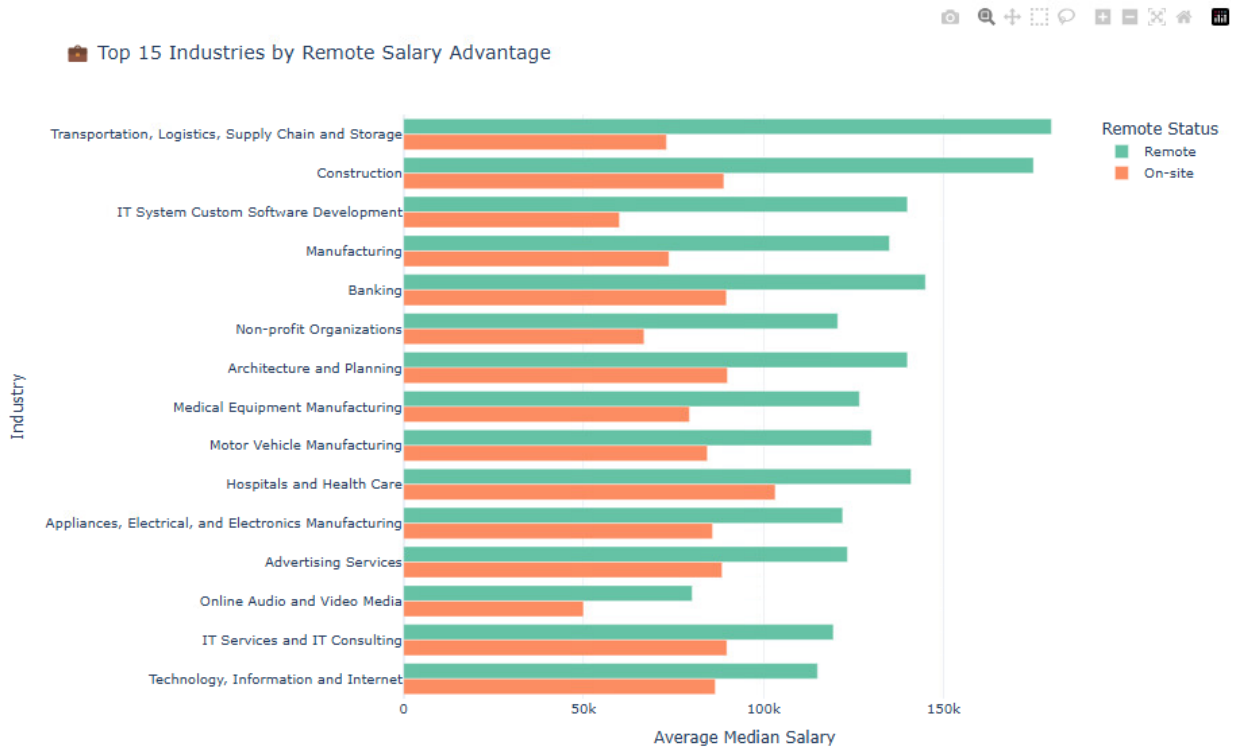
- **exp_level dominates** all other features with both high spread and impact, proving it's the **most predictive variable** in the salary model.
- **industry_code** and **title_code** follow next — validating that **what you do and where you do it** are key drivers.
- Features like remote_allowed and views have smaller, yet still consistent impacts suggesting **job visibility and remote flexibility do matter**, just less than core traits.
- ***Insight:*** For salary prediction, focus on **level, industry, and job title** — external factors like remote status help, but only marginally.

2. **Remote Work Insights:**

   Remote jobs showed a **significant salary premium** compared to on-site roles. The difference was statistically significant ($p < 0.0001$). This trend was especially strong in **tech, finance, and knowledge-based industries**, where remote roles commanded higher compensation.



- **Violin Plot**: Shows distribution of salary by remote status

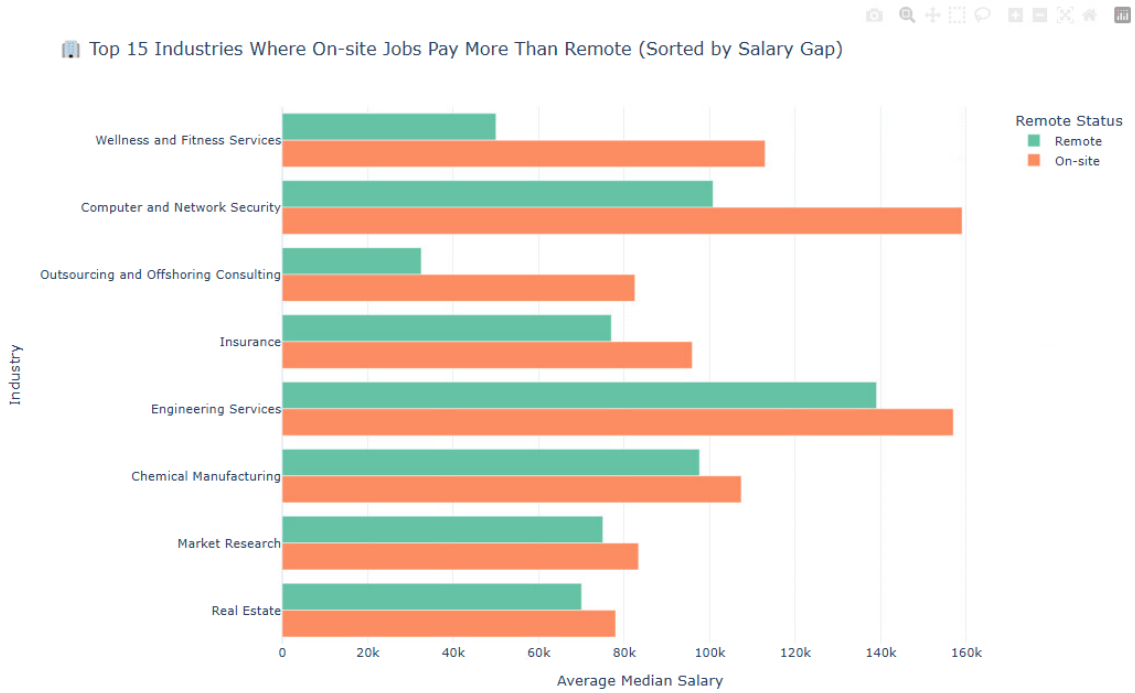


Bar chart comparing on-site vs remote salary across industries (e.g. Logistics, Banking, IT, Manufacturing).

Outliers: Industries Where Remote Pays Less

Industry	On-site	Remote
Outsourcing and Offshoring Consulting	$82,500	$32,505
Wellness and Fitness Services	$112,944	$50,000
Real Estate	$77,960	$70,000

Insight: These are service-based or hands-on roles where remote work may not be as lucrative or feasible.



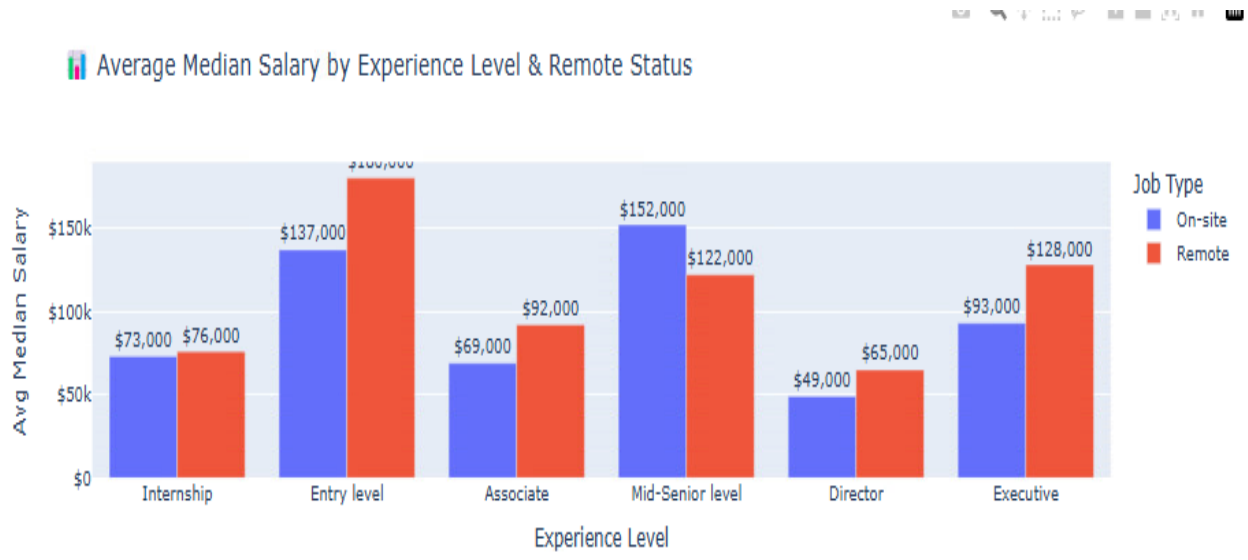
<u>Key Finding:</u>

At every experience level, remote jobs offer higher average salaries.

Experience Level	On-site Salary	Remote Salary
0	$73,002	$75,803
1	$137,464	$180,383
2	$69,669	$92,250
4	$49,827	$65,000
5	$93,799	$128,211

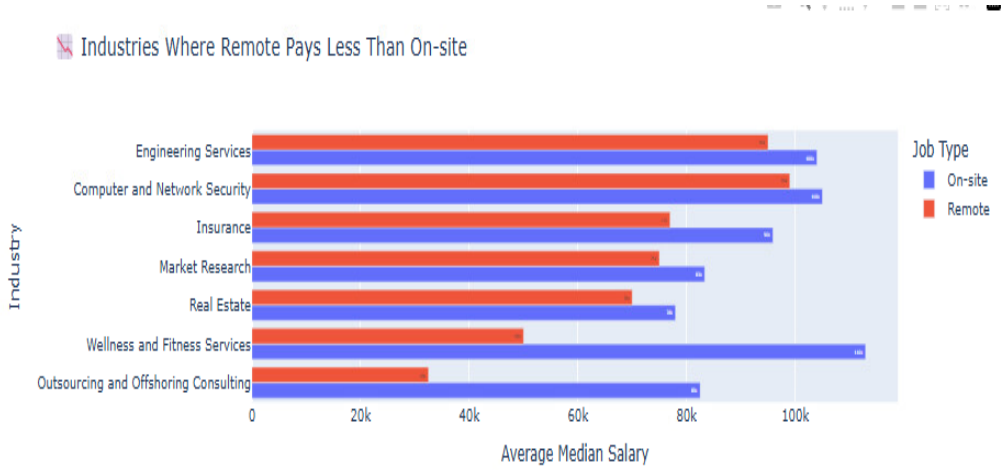
Visualizations:

Bar chart comparison by exp level



Side-by-side bar charts:

- Where remote pays more by industry
- Where remote pays less by industry
- Where remote pays more by experience (no loss cases)

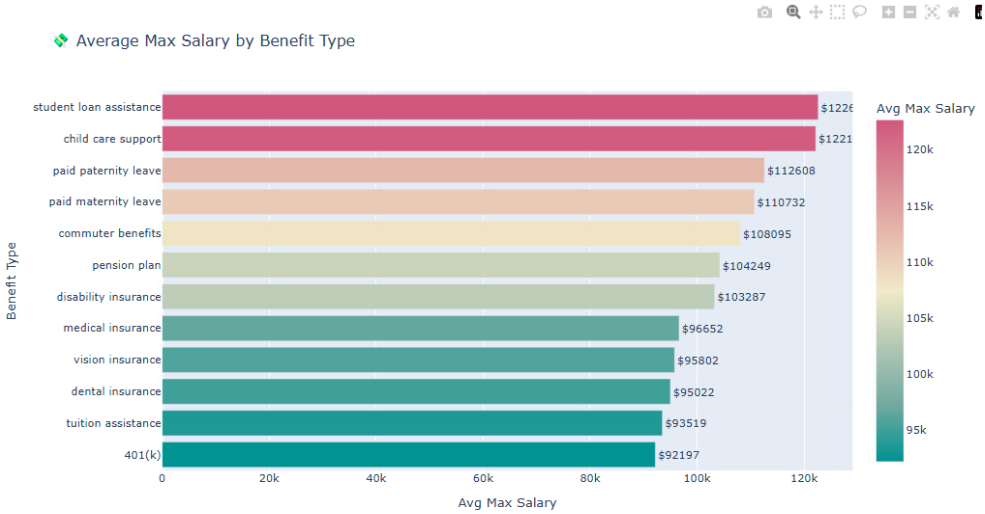


Remote Work Trends

- Most remote roles **offer salary premium**, especially in tech and knowledge-driven industries.
- Companies in **Fitness, Consulting, Insurance** still favor on-site pay advantage

3. **Benefits Analysis (401k):**

   Surprisingly, **jobs offering a 401(k)** did not consistently offer higher salaries. Its impact varied by industry and experience level — for example, **Directors and Interns** showed a benefit, while **Executives did not**. **Student loan assistance and child care** were associated with the highest average max salaries.



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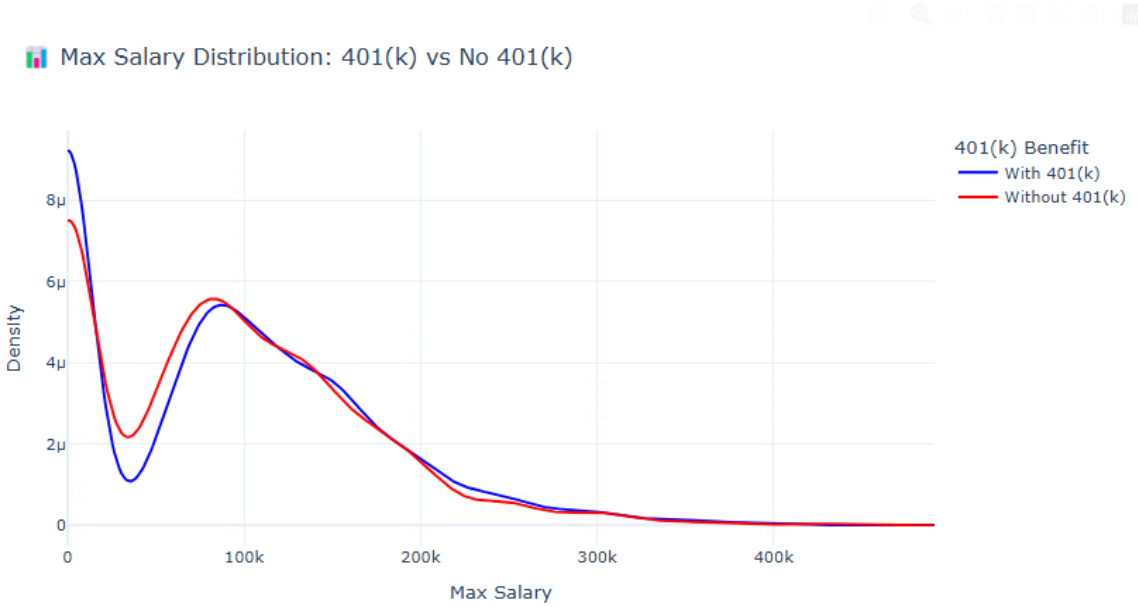
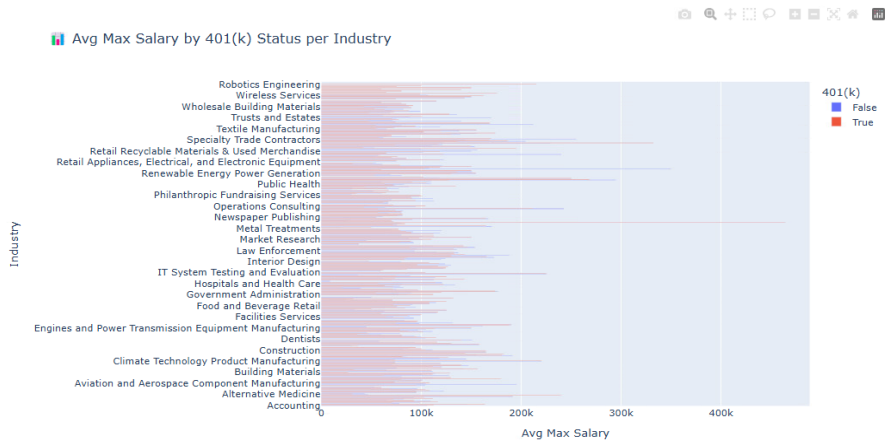
401(k) Benefit - Salary Distribution (Overall)

We compared the salary distributions between job postings with and without the 401(k) benefit.

- Initial violin plots and density plots showed overlapping distributions.
- **Improved KDE plot** finally revealed distinguishable distribution shapes.

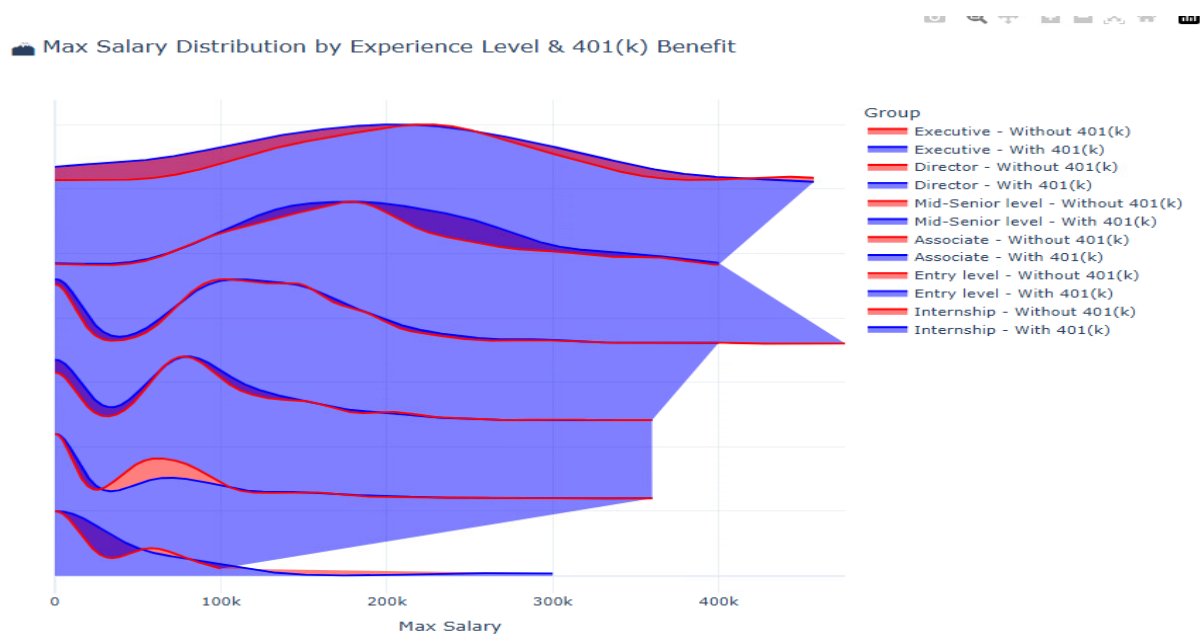




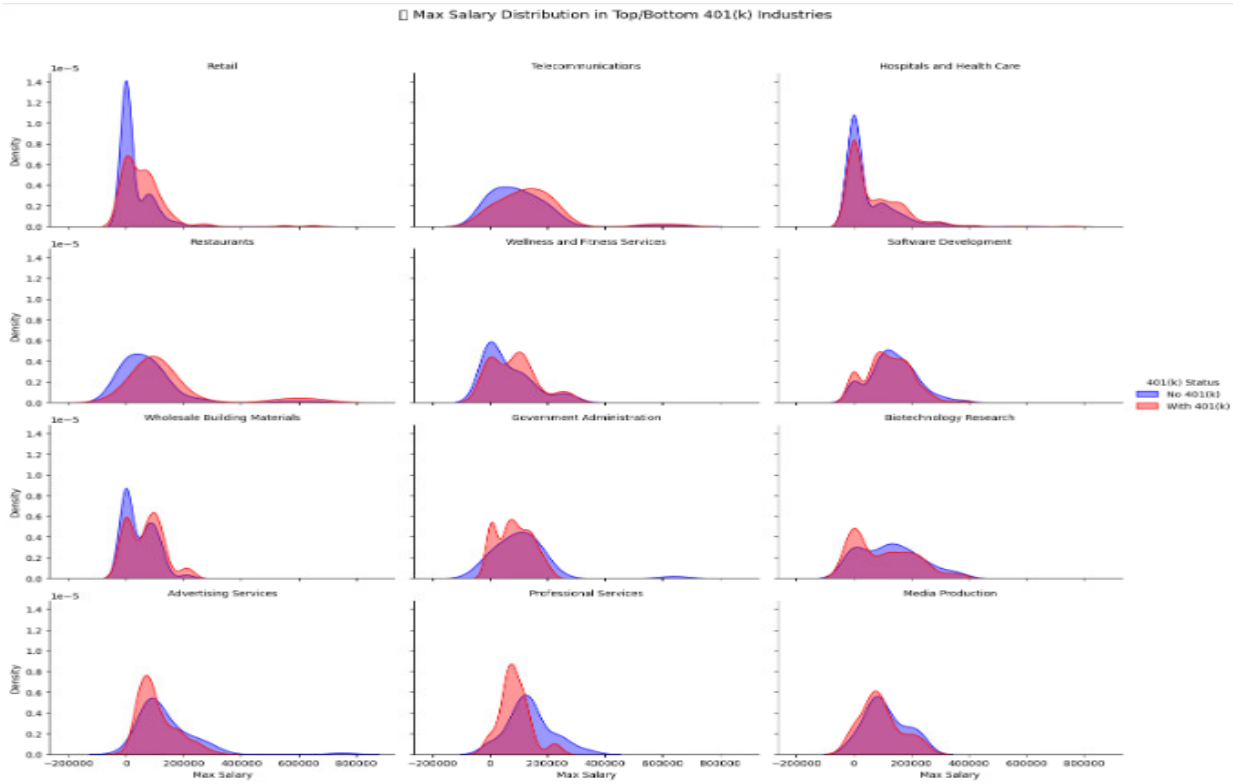
Ridgeline by Experience Level

We created a ridgeline plot showing the salary distributions by experience level, separated by 401(k) benefit status.

<u>Insight:</u> Most levels show higher salary density when 401(k) is included, especially for **Directors**.



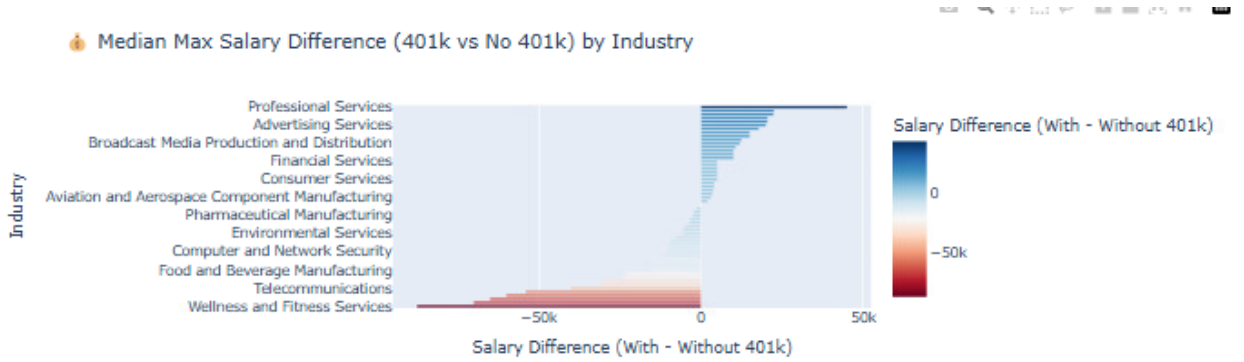
Comparative KDEs by Industry: For selected industries, KDE plots were plotted for with vs. without 401(k) status.

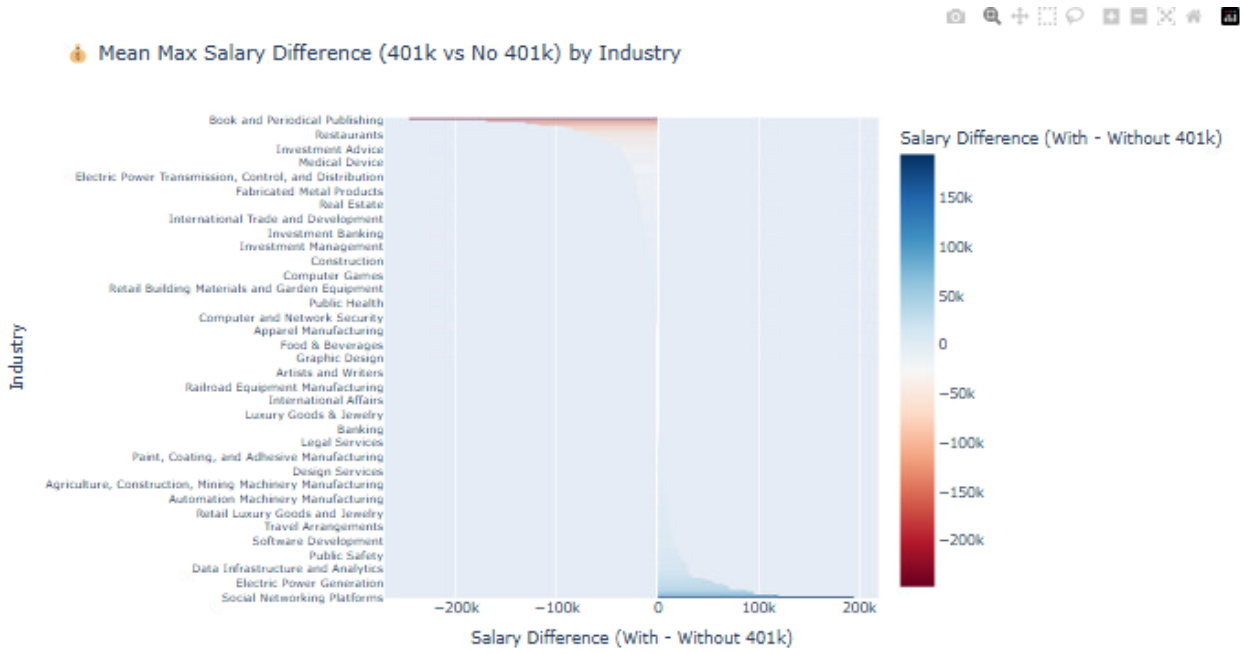


Mean Salary Difference by Industry

We visualized the average difference in max salary for jobs with vs. without 401(k), across industries.

- Industries with largest **positive** impact: e.g., *Advertising, Financial Services, Professional Services*.
- Industries with **negative** or neutral effect: *Telecom, Wellness, Retail*.

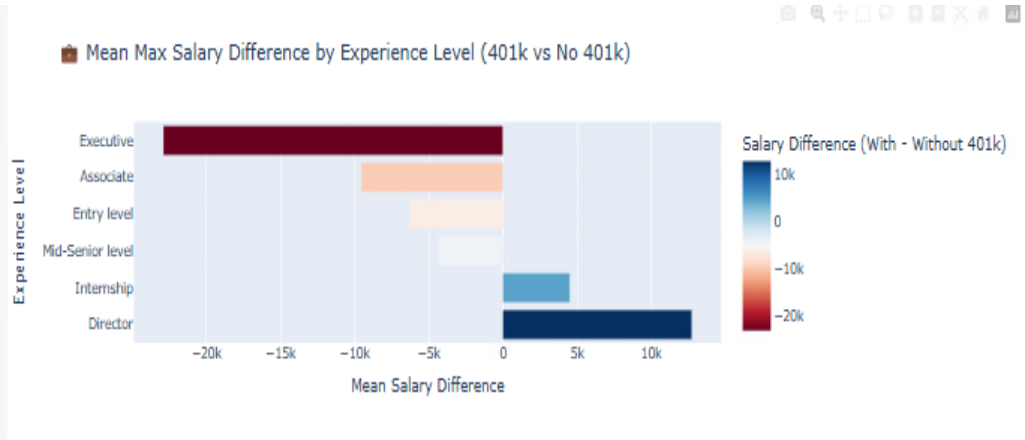




Mean Salary Difference by Experience Level

Finally, we summarized the impact of 401(k) on salaries across experience levels.

- **Directors & Interns** benefit most from 401(k) postings.
- **Executives** showed a **negative correlation**, possibly due to different compensation structures.



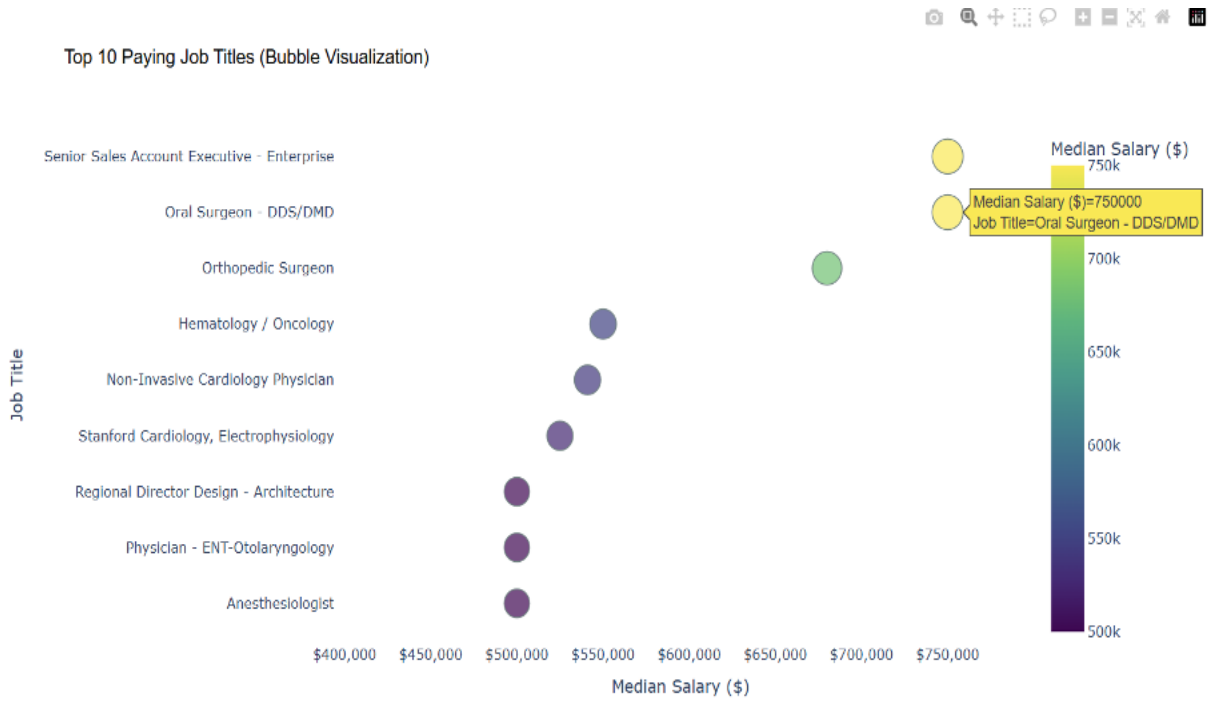
Insights:

- Having a **401(k) benefit** does not guarantee higher salaries.
- **Industry and experience level** play a significant role in moderating the salary benefit of a 401(k).
- **Mean-based measures** are more sensitive to outliers; medians might provide more stable insights.

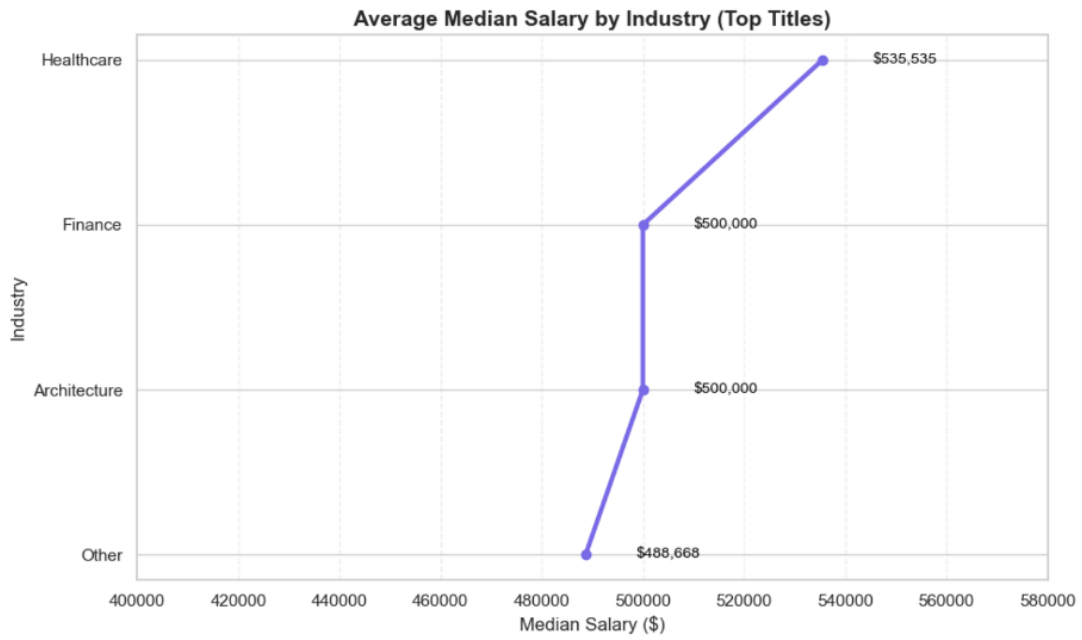
4. **Salary Transparency & Engagement:**
   In this analysis, we investigated the relationship between salary transparency and job listing engagement. Specifically, we examined whether companies provided salary details (min_salary, max_salary, or med_salary) and how that correlated with user engagement metrics like application count. Surprisingly, less than 5% of all job listings included any form of salary information, highlighting a widespread lack of transparency. When grouped by industry, major sectors such as Staffing and Recruiting, Software Development, and Retail showed 0% disclosure in posted salary data. This significant gap in transparency not only limits informed decision-making for job seekers but also hinders comparative analysis across roles and sectors. Our findings emphasize the need for more open compensation practices to foster trust and improve engagement.

5. **Top Paying Titles & Industries:**

   Filtering out internships, the **top-paying roles** were mostly in the **healthcare sector** (e.g., Orthopedic Surgeon, Oral Surgeon) with salaries reaching **$500K–$750K**. Executive roles also ranked high.



- The highest median salary was seen for **Oral Surgeon – DDS/DMD**, reaching **$750,000**, followed closely by **Orthopedic Surgeon** and other specialized medical roles.
- Most of the top-paying roles are in **healthcare**, particularly in highly skilled areas like **cardiology, oncology, and anesthesiology**.
- Despite variation in job title length or visibility, the **bubble size** indicates consistent high pay across top medical and technical fields.

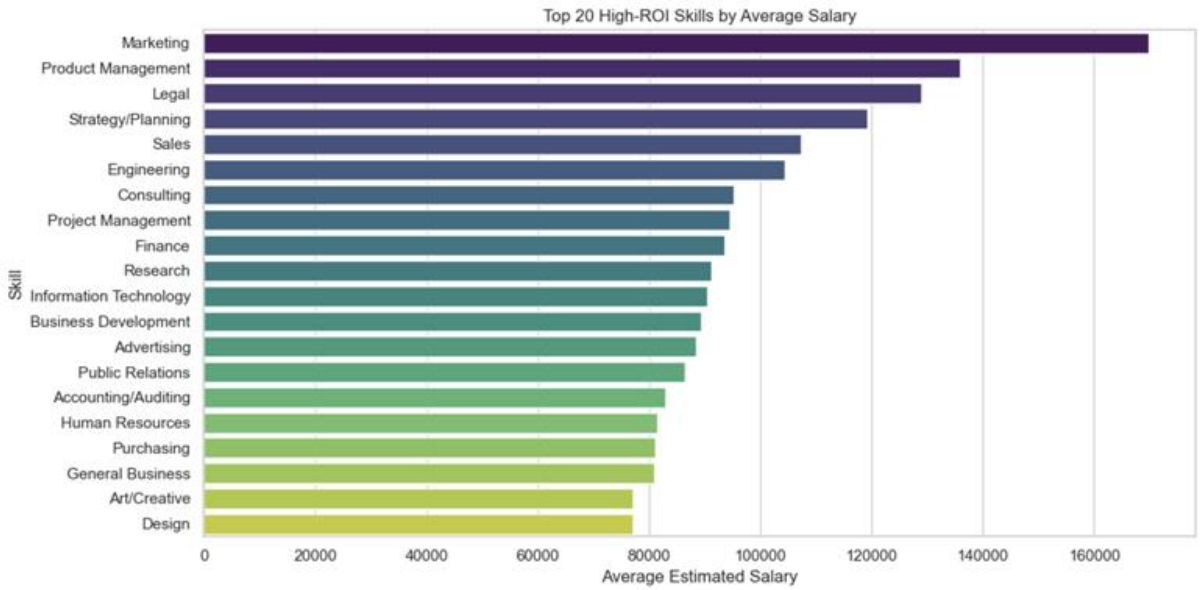


- **Healthcare** stands out with the **highest average median salary** across top roles, at **$535,535**.
- Industries like **Finance** and **Architecture** follow with competitive salaries around **$500,000**, showing the value of specialized knowledge and leadership in these sectors.
- The **"Other"** category has slightly lower pay, possibly due to more generalized or varied roles.

Takeaway:

Top-paying job roles are heavily **concentrated in healthcare**, both in absolute terms and industry averages. Professionals with advanced degrees and niche specialties are rewarded the most. Finance and design leadership roles also offer strong compensation but are outpaced by medical professions in terms of median salary.

6. **High ROI Skills:**
   Skills like **Marketing**, **Product Management**, and **Legal** offered the highest average salaries. Combining **technical (e.g., Engineering)** and **business skills (e.g., Strategy)** yielded the best pay outcomes. **Design** and **Art** skills ranked lower.



<u>Insights:</u>

- **Marketing, Product Management, and Legal** emerged as the top skills associated with the highest average salaries — all exceeding $130,000, indicating strong demand in strategic and leadership roles.
- Skills like **Engineering**, **Consulting**, and **Strategy/Planning** also showed consistently high returns, highlighting the value of combining technical depth with business acumen.
- On the other end, creative and generalist roles like **Design** and **Art/Creative** had lower average salaries, suggesting they may be undervalued in the current job market.
- The **Word Cloud** reinforces these trends visually, with the most lucrative and frequently mentioned skills like *Marketing* and *Product Management* shown in larger fonts.
- Overall, this analysis provides a practical benchmark for professionals to **prioritize high-paying skillsets**, and for employers to **strategically align job descriptions** with market-valued expertise.

7. **Certification Signals:**
   Simply mentioning "AWS Certified" had misleading salary patterns due to small sample size and recruiter phrasing. Using broader terms like **"AWS"** gave more reliable insights. This highlighted how **text mining sensitivity** affects analysis.

   We ran the initial analysis and plotted the hourly median salary comparison, something strange stood out: **jobs that mentioned "AWS Certified" had lower pay than those that didn't.**

This didn't sit right with us. We brainstormed possible reasons:

- **Very small sample size**: Only a few job descriptions actually mentioned "AWS Certified."
- **Lower-level roles might mention it more directly**: Support and admin jobs tend to include certs explicitly.
- **Buzzword filler?** Recruiters may add "AWS Certified" without it really being part of a high-paying position.

But then it hit us; maybe it's not the certification that's skewing the results, but how we were searching for it. The phrase **"AWS Certified"** is very specific. If some descriptions just said, **"AWS experience"** or **"AWS cloud"**, we wouldn't catch them. And worse, if we only search for **"AWS Certified"**, the few results we're getting might be misrepresentative, which were only 3.
So, we changed our approach: we simplified the keyword to just **" AWS "** (with spaces to avoid words like "saws" or "lawson"). This captured **any AWS-related skill or tool**, giving us more rows and a more reliable median.
After rerunning the analysis, **the median salary jumped to where we expected it to be; above** non-certified listings.

This was a good reminder that **text mining is sensitive to phrasing**, and small tweaks in keyword logic can change insights dramatically.

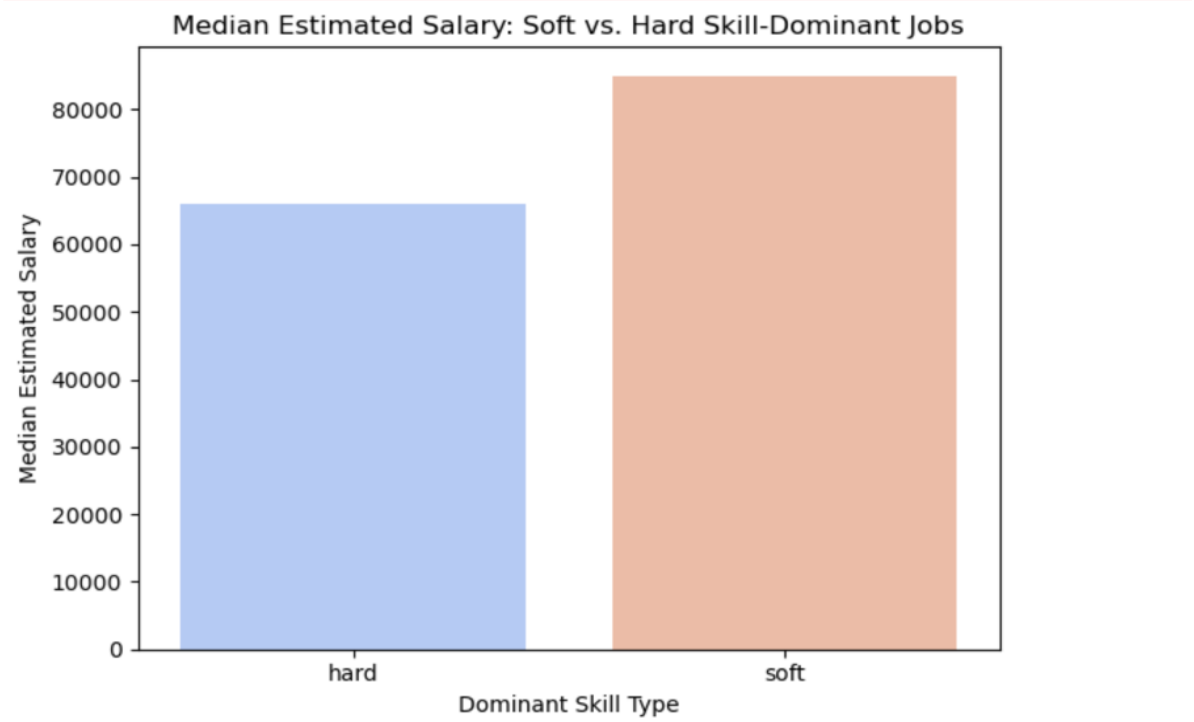
<u>Takeaway:</u>

Certified professionals consistently earn higher hourly wages, especially in fields like cloud computing, finance, and project management. Credentials like CPA, Google Cloud, and PMP show strong ROI, making them valuable for boosting compensation.

8. **Soft vs. Hard Skills:**
   Jobs dominated by **soft skills** (e.g., leadership, communication) had a **median salary of $85K**, outperforming **hard skill** jobs (e.g., coding, SQL) at **$65K**. This suggests leadership and strategy roles drive higher value.

```
Median Estimated Salary by Dominant Skill Type:

dominant_type
hard    66000.0
soft    85000.0
Name: estimated_salary, dtype: float64
```

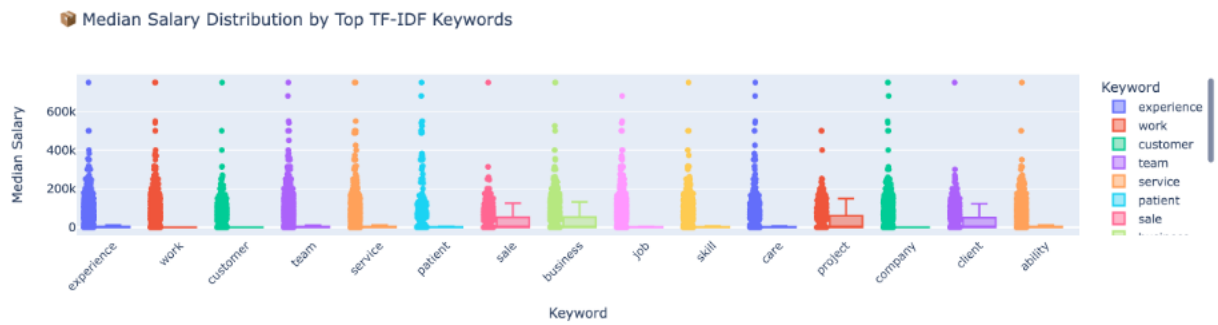


Insights:

- **Marketing, Product Management, and Legal** top the chart as the highest ROI skills, offering average salaries above $130K, signaling strong demand in business, strategic, and compliance-driven roles.
- The **word cloud** further supports this by emphasizing the frequency and prominence of skills like *Strategy/Planning*, *Consulting*, *Engineering*, and *Sales*, which appear both often and with competitive pay.
- When comparing **soft vs. hard skills**, roles dominated by soft skills (e.g., leadership, communication, collaboration) had a **median salary of $85K**, while those with hard technical skills (e.g., Python, SQL, Excel) earned **$66K**.
- This suggests that **skills tied to decision-making, people management, and business influence** drive higher pay than purely execution-focused technical skills.
- **Key takeaway**: While technical skills get you in the door, combining them with soft skills (like strategic thinking or leadership) is what **amplifies earning potential** in today's job market.

9. **Keyword-based Salary Prediction:**

We used TF-IDF to extract the most common keywords from job descriptions and analyzed how they relate to median salary. The chart shows salary distributions for each keyword, helping us understand if certain terms are tied to higher or lower-paying roles. While many keywords were generic and appeared across various job types, some showed patterns linked to specific industries or pay ranges. This approach offers a text-based perspective on how job language may reflect salary trends.



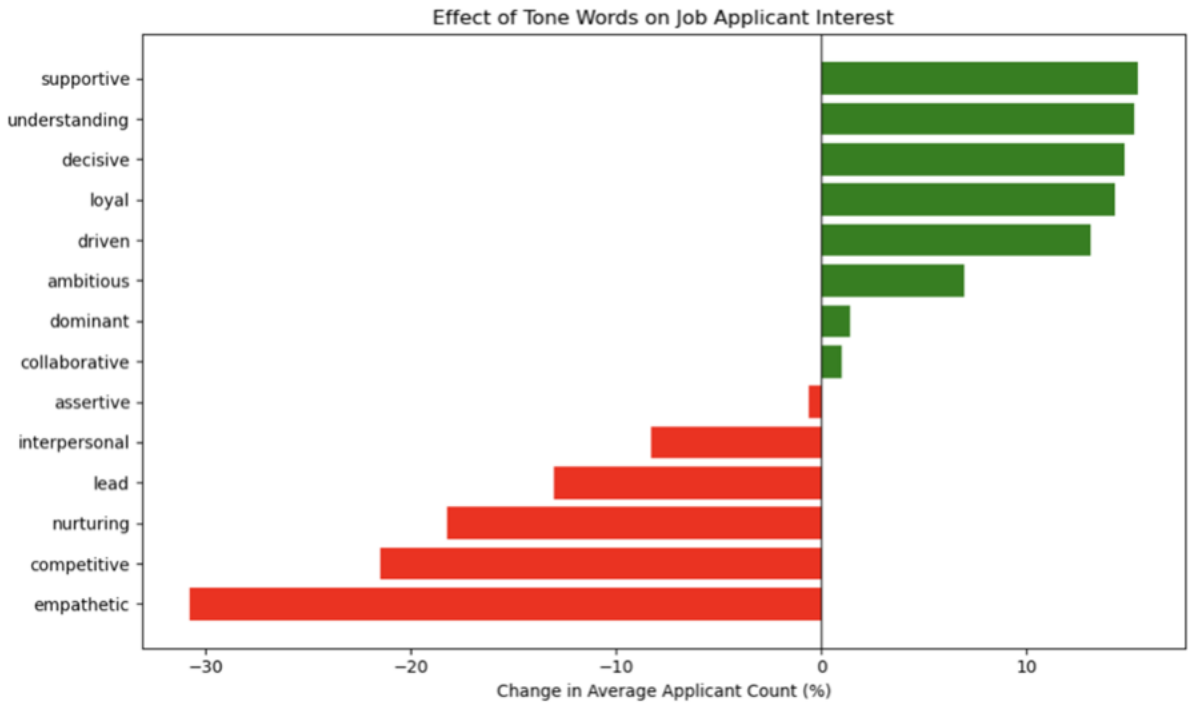
The chart shows median salary distributions for top keywords found in job descriptions using TF-IDF, helping us explore how certain terms relate to salary levels.

<u>Takeaways:</u>

- **Wide Salary Spread:**
  Most keywords (like "experience", "team") appear across all job levels, leading to large salary ranges and many outliers.
- **Generic Terms Dominate:**
  Common words such as "work", "company", and "skill" show little predictive power alone, as they're found in many types of roles.
- **Role-Specific Clues:**
  Keywords like "client", "care", and "patient" are likely tied to healthcare/support jobs, showing more consistent, lower pay ranges.
- **High-Paying Outliers:**
  Words like "project" and "business" have visible outliers, hinting at executive or senior-level roles.
  <u>Insight:</u> While these keywords offer some context, they're best used alongside structured variables for accurate salary prediction.

10. Gendered Language in Job Descriptions:

Using tone-coded keyword analysis from social science research, we found that **feminine-coded words** like *"supportive," "understanding," "loyal,"* led to **15%+ more applicants**, while assertive terms like *"competitive"* and *"lead"* reduced interest. Most high-performing words fell into the **feminine-coded category**, showing that **inclusive, empathetic language increases applicant engagement**. This insight gives companies a low-cost tool to **broaden applicant pools and improve diversity**.



This chart analyzes the effect of tone-based words on job applicant interest. Words like *supportive*, *understanding*, and *loyal*—typically categorized as feminine-coded—led to a 15%+ increase in applicant count. In contrast, assertive terms like *competitive*, *lead*, and *empathetic* saw significant drops in interest. The findings highlight how inclusive and empathetic language can positively influence applicant engagement.

<u>Takeaway:</u>

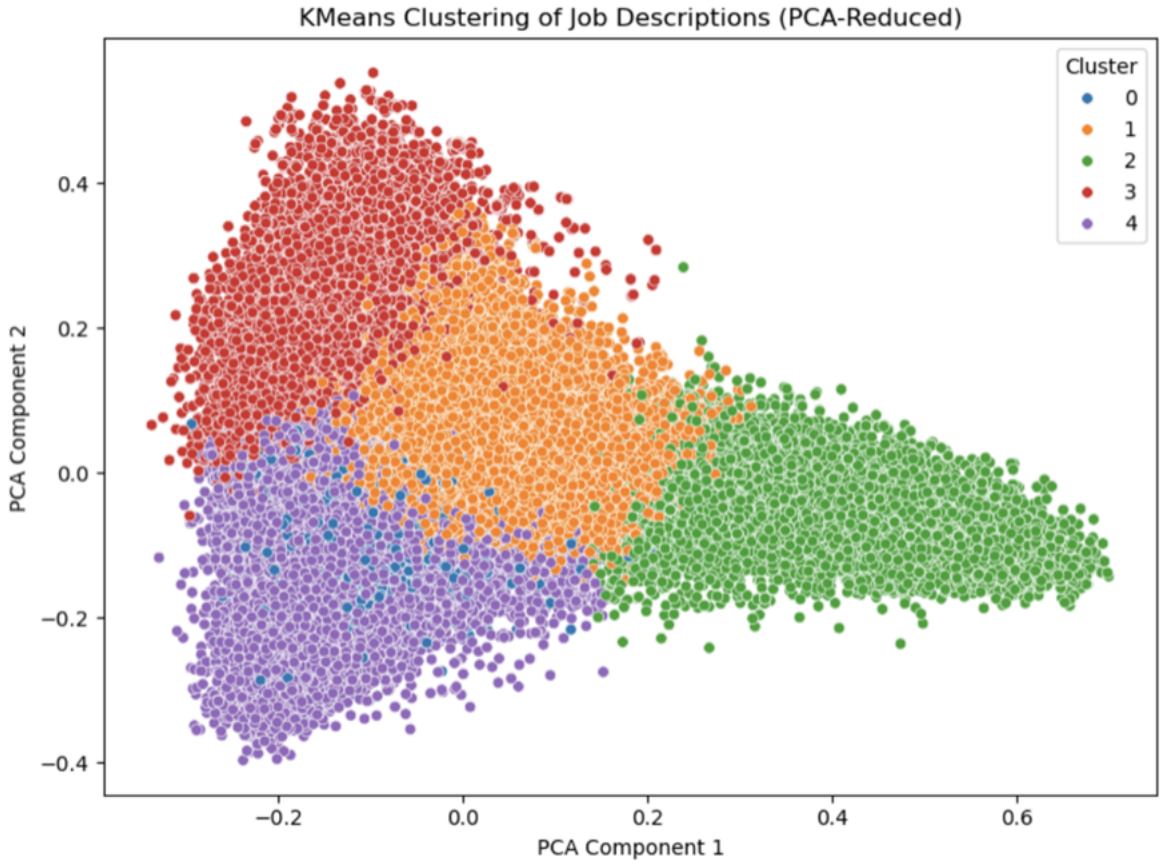
Inclusive tone words like *supportive* and *understanding* significantly boost applicant interest. In contrast, assertive terms such as *competitive* and *empathetic* may discourage applicants.

11. Clustering Job Descriptions:

We used **KMeans clustering** on **TF-IDF vectors** of job descriptions and visualized the results with **PCA**, revealing 5 distinct clusters.

- **Cluster 0**: Finance/Admin (e.g., FP&A Analyst) – median $26/hr
- **Cluster 1**: Event-based/Part-time (e.g., Intern, Coach) – lowest pay at $10.88/hr
- **Cluster 2**: Clinical Healthcare (e.g., RN, PA) – highest at $55/hr
- **Cluster 3**: Retail/Sales (e.g., Customer Service) – $21/hr
- **Cluster 4**: Hybrid Professional (e.g., Attorney, Marketing Manager) – $65/hr

These clusters confirmed known role structures and uncovered **hybrid zones** with unique pay ranges. Companies can use this method to detect **role overlaps**, guide **strategic hiring**, and identify **salary inconsistencies** across teams.



We found this clustering useful not only for confirming intuition (like Cluster 2 being healthcare jobs), but also for revealing **hybrid zones**, like Cluster 4 — jobs that blend technical, legal, or creative responsibilities.

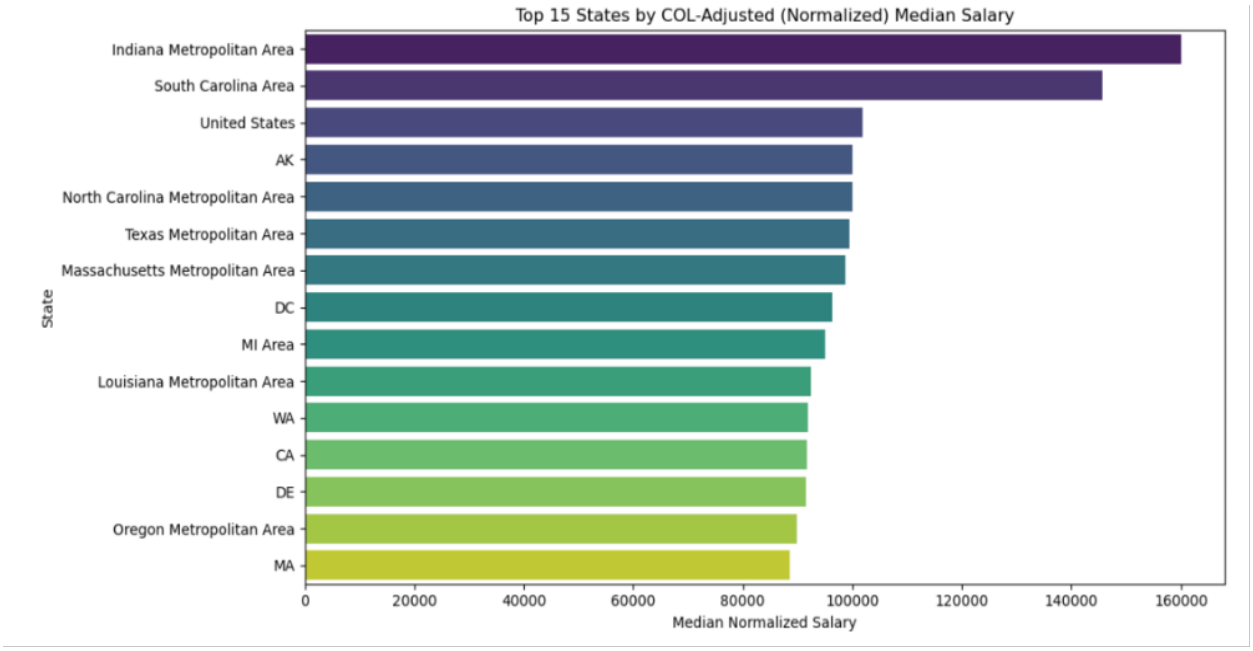
Another thing that stood out was how salary patterns validated the clusters. Clusters 1 and 3 — which mostly include temporary or sales support roles — had the **lowest hourly pay**, while Clusters 2 and 4 — healthcare and specialized professionals — had **the highest**. This made us realize that if a company is analyzing their job catalog, they could use this kind of clustering to **identify overlap, hybrid roles, or even spot underpaid positions relative to their peers** — especially for strategic workforce planning.

Takeaway:

Inclusive tone words like *supportive* and *understanding* significantly boost applicant interest. In contrast, assertive terms such as *competitive* and *empathetic* may discourage applicants.

12. Cost of Living Normalization:

We performed cost of living (COL) normalization using the normalized_salary field from the cleaned job postings dataset. The data was filtered to exclude missing values and extreme salary outliers. We then extracted state names from the location column and grouped the data by state. This allowed us to compare median COL-adjusted salaries across different regions and visualize which states offer better real income value.



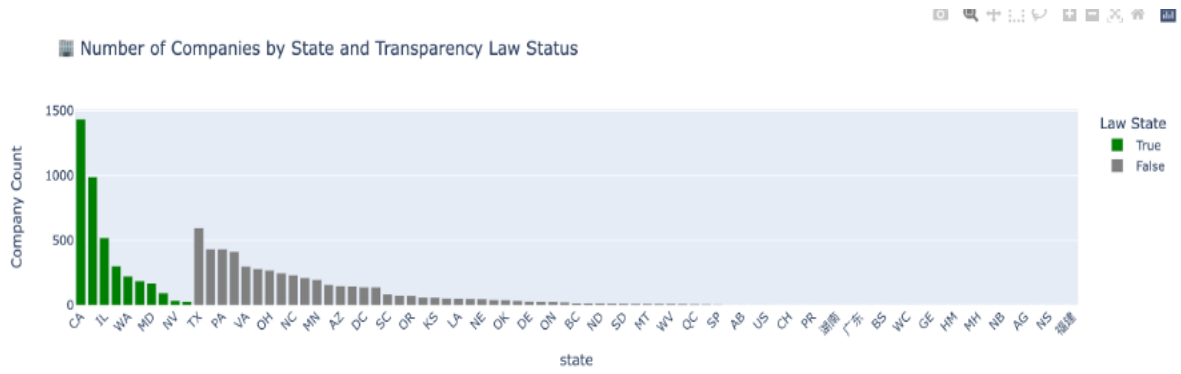
Insights:

- **Indiana** and **South Carolina** metropolitan areas offer the **highest real income value**, with COL-adjusted median salaries exceeding **$140K–$150K**, making them highly attractive for remote or cost-savvy job seekers.
- Surprisingly, less expected regions like **Alaska** and **North Carolina** also rank high, challenging the idea that only big tech hubs offer top earnings.
- States like **California** and **Massachusetts**, despite offering high nominal salaries, rank lower after COL adjustment — showing that **living expenses significantly impact true earning power**.
- The inclusion of generic labels like *"United States"* or *"Metropolitan Area"* in the rankings suggests some **location standardization or cleanup** might improve analytical precision.

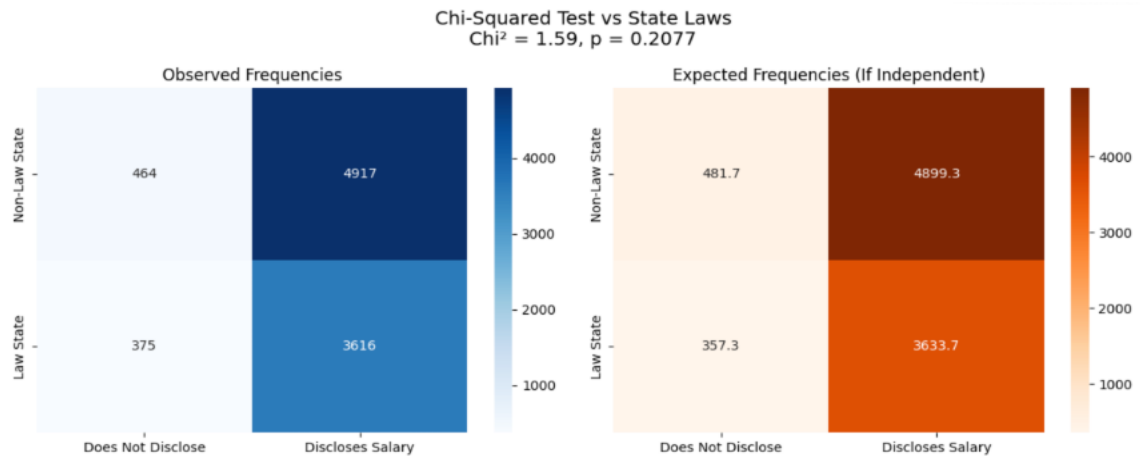
Takeaway: When job hunting across states, **don't just chase big numbers — normalize for cost of living** to identify where your salary truly stretches the furthest.

13. Transparency by State:

We explored salary transparency across U.S. states by combining company headquarters data with each state's salary disclosure laws. We compared company counts across law and non-law states and visualized disclosure patterns using bar charts. To test for statistical significance, we performed a **Chi-squared test** on salary posting behavior across law vs. non-law states. This helped us evaluate whether transparency laws actually impact disclosure rates in job listings.



- CA leads with the highest number of companies.
- Law states like NY, WA, IL also show strong company presence.
- TX has high company count without a transparency law.
- Mentions malformed/international entries as a data cleaning consideration (briefly implied by platform norms mention).



- Observed and expected frequencies of disclosure are very close.
- Clear note that **p = 0.2077**, not statistically significant.
- States that **disclosure behavior isn't driven solely by law presence**—instead, company/platform culture may influence it more.

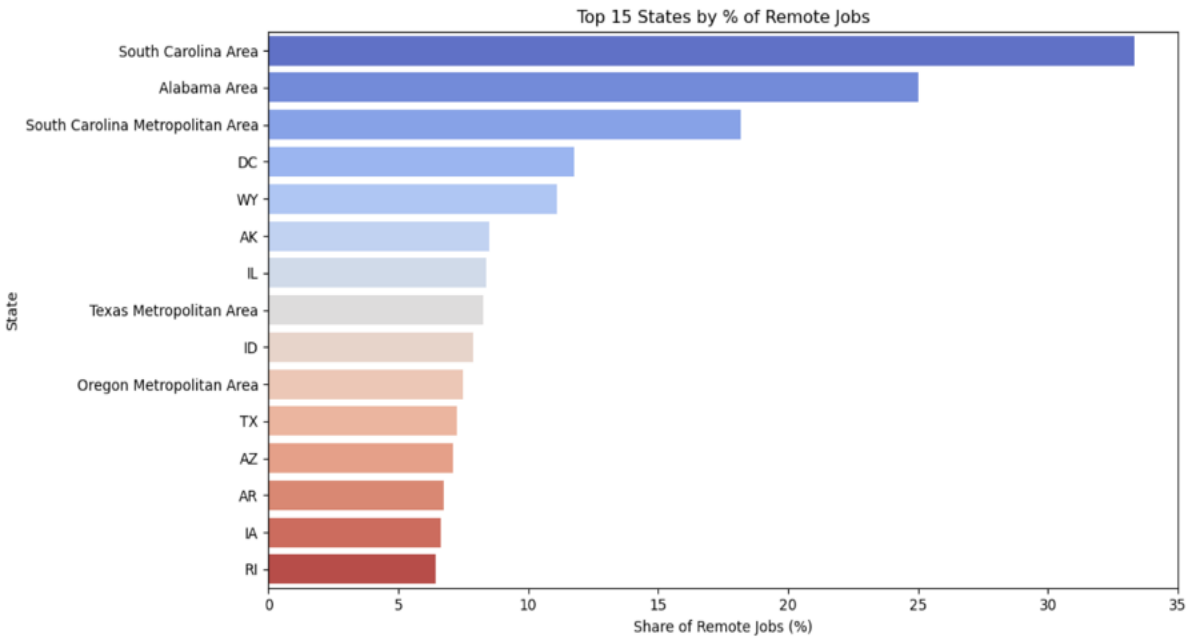
<u>Takeaway:</u>

- Transparency laws may signal best practices, but **enforcement and culture play a bigger role** in actual salary disclosure.
- Companies in **non-law states** still frequently disclose salaries, especially in competitive or tech-driven markets.
- **Policy presence alone doesn't guarantee behavior change**—internal values and industry standards may matter more.

14. Remote Job Distribution:

We analyzed remote job availability across U.S. states using cleaned_postings.csv, focusing on jobs with valid remote_allowed and normalized_salary values. After

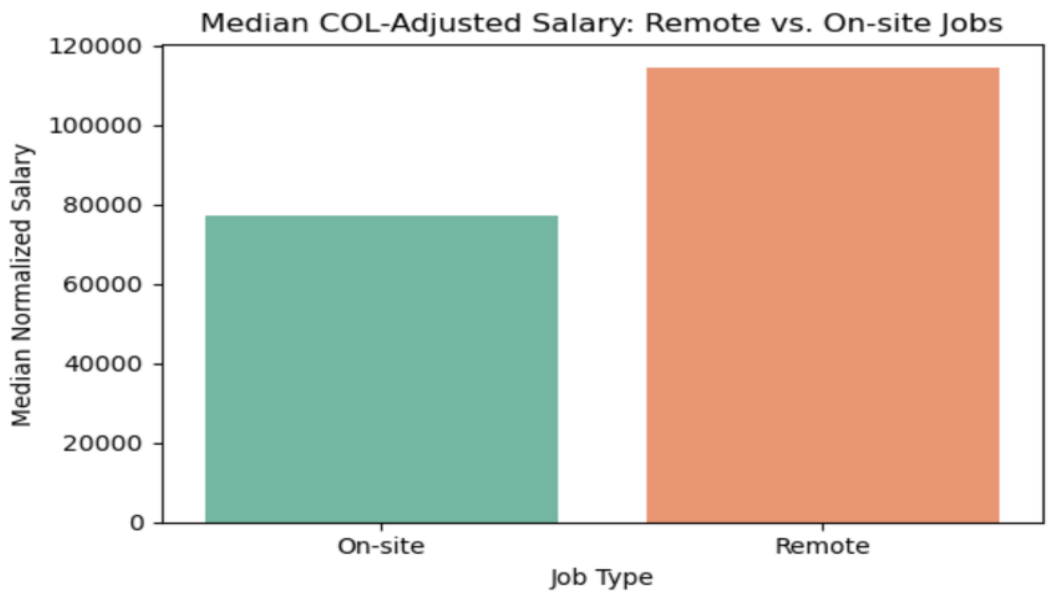
extracting and cleaning state information from the location column, we calculated the share of remote jobs per state. This allowed us to identify regions with higher remote work adoption. We then visualized the top states by percentage of remote job postings.



- **Indiana** and **South Carolina** lead the chart, with normalized median salaries exceeding $140,000.
- Surprisingly, less prominent states like **Alaska** and **North Carolina** also offer strong purchasing power.
- High nominal salary states like **California (CA)** and **Massachusetts (MA)** rank lower due to high living expenses reducing the actual value of earnings.

Insight:

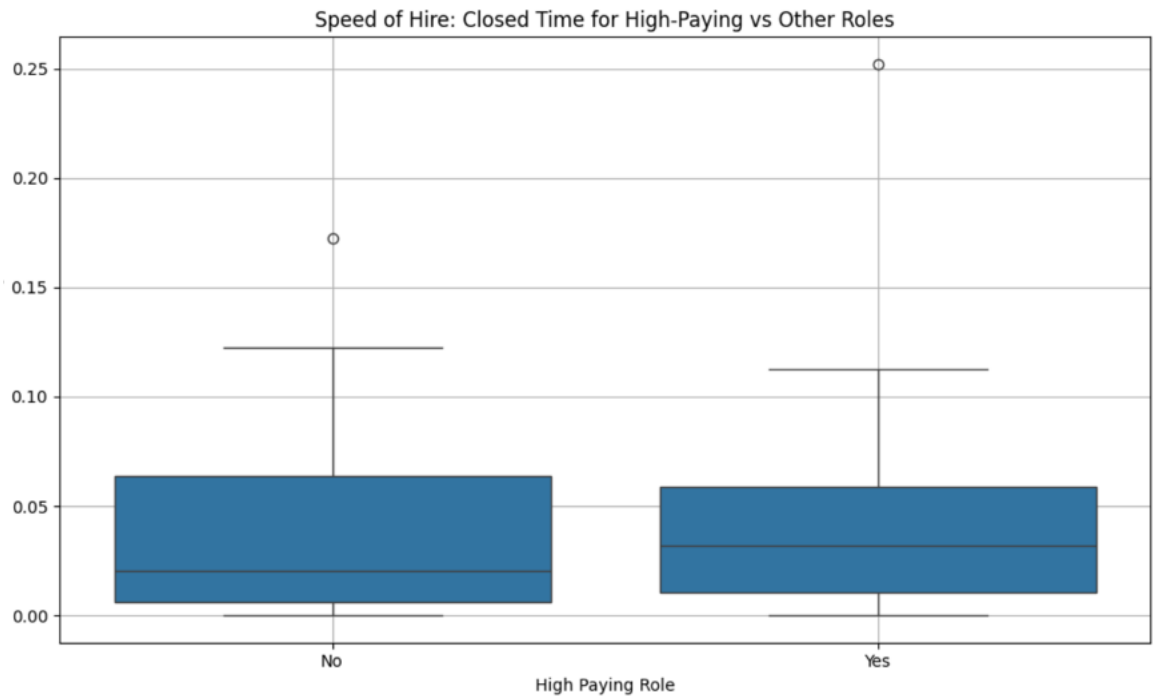
Candidates can maximize their real income by targeting jobs in lower-cost, high-salary states — making location a crucial factor in job selection.



- **Remote jobs offer a median normalized salary of ~$115K**, significantly higher than **~$77K for on-site jobs**.
- This suggests that companies are willing to pay more for remote flexibility, especially in high-demand fields like tech and consulting.

Insight:
Remote work not only provides lifestyle flexibility but also offers a tangible salary advantage — especially for individuals living in affordable regions.

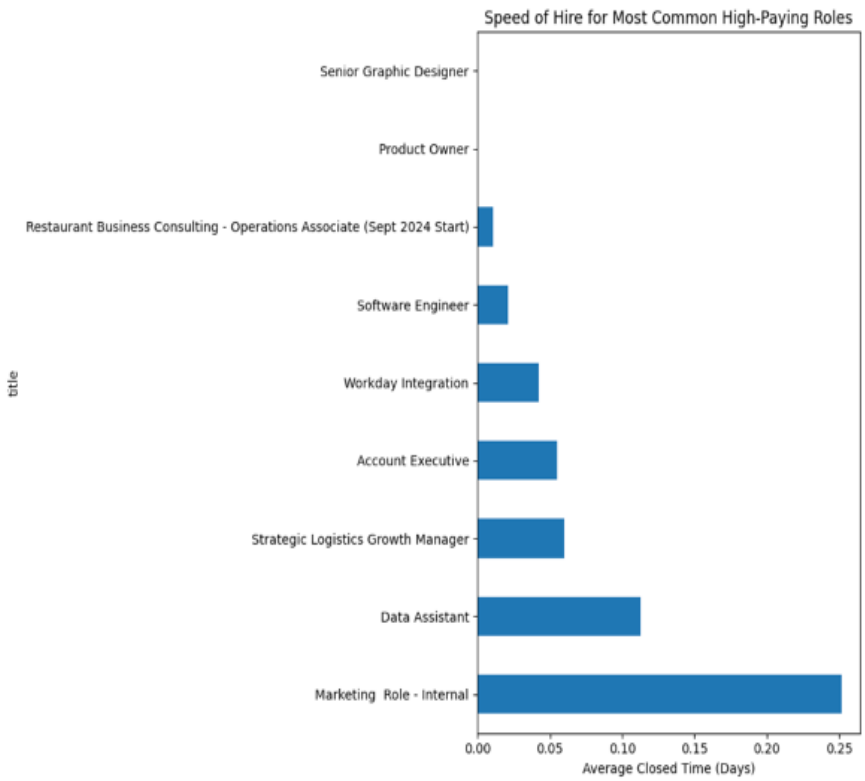
15. **Speed of Hire:**
We analyzed how quickly high-paying roles are filled by calculating the closed_time between job listing and removal. After normalizing salaries and identifying the top 25% as high-paying, we computed average closing durations by job title. Time columns were converted from Unix format for accuracy. This helped identify roles that close especially fast — potentially due to high demand or internal hiring practices. The findings were visualized through boxplots and bar charts to reveal hiring speed trends across top-paying positions.



This graph compares the distribution of closed times (in days) for high-paying vs non-high-paying roles.

Insights:

- **Median closed time is slightly lower** for high-paying roles, suggesting they are generally filled faster.
- **Most roles in both categories are closed in under 0.05 days**, indicating listings are often taken down quickly.
- The presence of **outliers** in both categories implies that some roles, regardless of salary, remain open longer due to niche requirements or delays.



This graph shows the **average closed time** (in days) for the top 10 most frequent high-paying job titles.

Insights:

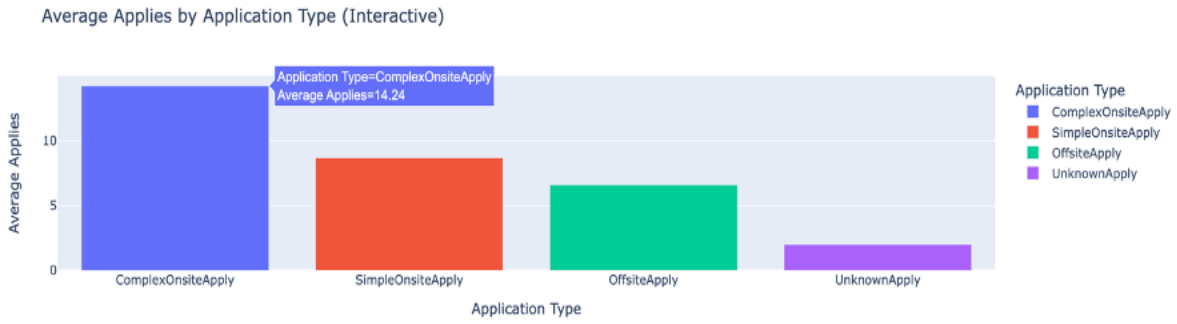
- **Fastest-filled roles:** *Senior Graphic Designer, Product Owner, Software Engineer* — often closed within a day, reflecting high demand and perhaps pre-identified candidates.
- **Slower to fill:** *Marketing Role – Internal, Data Assistant* took more time, likely due to extended evaluation or internal hiring processes.
- Despite being high-paying, these roles vary in speed—**popularity and internal processes** influence closure.

Takeaway:

High-paying jobs are often filled rapidly, especially in tech and design roles, due to high demand. However, average closed times still vary across titles—suggesting that salary isn't the only factor influencing hiring speed.

16. Quick Apply vs. Apply Rate:

We explored how different application_type formats affect the number of applies a job receives. By grouping postings and computing average applies by type, we found that **ComplexOnsiteApply** surprisingly had the highest engagement. This challenges assumptions that simpler workflows always lead to more applications. The findings suggest job quality and visibility may outweigh the ease of application in driving applicant interest. Results were visualized using bar charts to highlight engagement trends across application types.



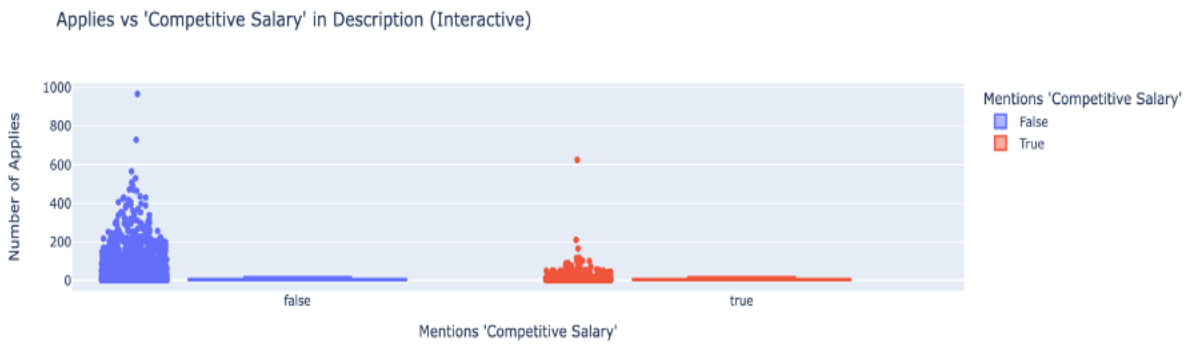
<u>Insights:</u>

- **ComplexOnsiteApply** has the **highest average applies (~14)**, suggesting that despite being more time-consuming, these jobs might be perceived as higher quality or better promoted.
- **SimpleOnsiteApply** receives a **moderate number of applies**, likely benefiting from ease of use but less company-driven promotion.
- **OffsiteApply** has lower engagement, possibly due to redirection or extra steps that introduce friction.
- **UnknownApply** sees the **least engagement**, likely due to lack of clarity or missing application metadata.

<u>Takeaway:</u> Ease of application doesn't always guarantee more applies—**job quality, promotion, and perceived value** also play critical roles in driving candidate interest.

17. Competitive Salary" vs Apply Rate:

We analyzed whether the phrase "competitive salary" in job descriptions affects the number of applications received. Jobs using this vague language had **lower average applies** and fewer high-apply outliers, as shown in the boxplot. This suggests applicants prefer **transparency over ambiguity** in salary communication. The presence of this phrase may signal withholding of important info, leading to reduced interest. Results reinforce the value of being explicit about compensation.



<u>Insights:</u>

- **Jobs without** the phrase *"competitive salary"* received **higher and more variable application counts**, with several outliers exceeding 500+ applies.
- **Jobs with** the phrase showed **lower apply rates** and almost no high-outlier spikes.
- This suggests that **vague language may discourage applicants**, likely due to **lack of transparency or perceived ambiguity** in compensation.

<u>Takeaway</u>: To boost candidate interest and trust, **explicit salary disclosure is more effective** than using generic terms like *"competitive salary"*.

18. Gender-Neutral Descriptions:

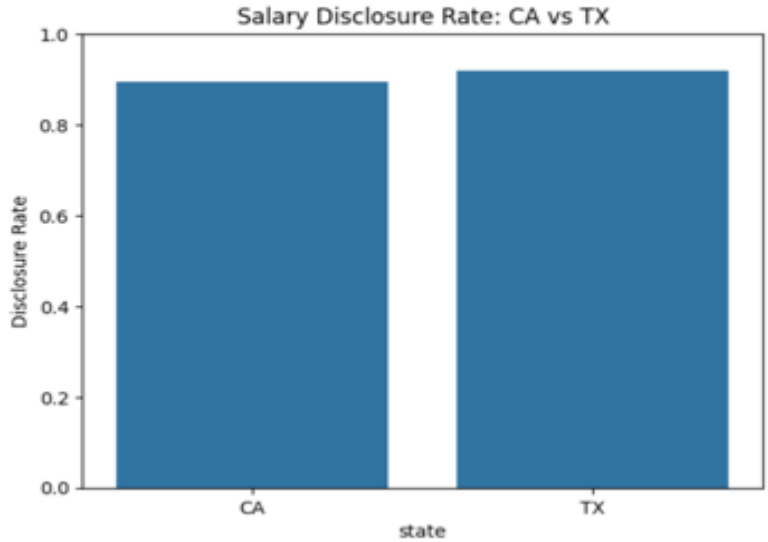
As previously analyzed in the ***Gendered Language in Job Descriptions* section**, we used tone-coded keyword analysis based on social science research to understand how language affects applicant behavior. Words coded as feminine — such as "supportive," "understanding," and "loyal" — were found to increase applicant interest by over 15%. In contrast, assertive or masculine-coded terms like "competitive" and "lead" were associated with reduced engagement. Although we couldn't assess salary bias directly due to missing applicant demographics, the analysis clearly shows that **inclusive language can broaden applicant pools** and serve as a practical lever for equitable hiring practices.

19. Salary Transparency by Region:

We compared salary disclosure rates in California (CA) and Texas (TX), using both bar charts and a chi-squared statistical test. Surprisingly, TX had a slightly higher transparency rate (91.9%) than CA (89.5%), despite CA having salary transparency laws. However, this difference was **not statistically significant** ($Chi^2 = 2.72$, $p = 0.09896$). This suggests that factors beyond legislation — such as company culture or platform policies — may influence transparency behavior.

```
        sum  count  disclosure_rate
state
CA     1285   1436         0.894847
TX      551    599         0.919866

Chi-squared Test: Chi² = 2.72, p = 0.09896
```



- Both California (CA) and Texas (TX) show high salary disclosure rates, above 89%.
- Surprisingly, TX (91.9%) slightly outperforms CA (89.5%), despite CA having formal transparency laws.
- However, the Chi-squared test result (p = 0.099) indicates this difference is not statistically significant.

  Insight: Transparency seems to be driven more by company culture or platform norms than by state law enforcement.

Predicted Salary by Location:

- **Top Locations**: Beeville (TX), Baldwin Park (CA), and Jasper (IN) show exceptionally high predicted salaries — possibly due to niche or executive roles.
- **Bottom Locations**: Places like Spiro (OK) and Elizabethtown (NY) have very low medians (~$11K), likely reflecting data issues or misclassified hourly jobs.

  Insight: Salary extremes often stem from either role-specific spikes or inconsistencies in salary normalization. Filtering low-posting locations is essential for fair comparison.

Takeaways:

- High transparency is visible in both regulated and non-regulated states — regulation alone doesn't explain behavior.
- Median salary can vary drastically by location; always cross-check with job types and posting volume for accuracy.

Discussion

Our analysis of LinkedIn job postings provided deep insights into labor market trends, employer strategies, and candidate behaviors. We observed that **experience level, job title, and industry** are primary drivers of compensation, while **soft skills** and **inclusive language** significantly influence engagement. **Remote jobs** not only pay more but also reflect shifting geographic preferences, and our **seasonality analysis** highlighted cyclical hiring patterns. Importantly, **salary transparency** and **application method** directly impact applicant volume. These findings empower job seekers to prioritize high-ROI skills and guide employers in crafting optimized, inclusive, and competitive job listings.

Limitations

- Approximately **70% of postings lacked normalized salary** values, which restricted comprehensive salary-based analyses and comparisons.
- **Job titles and benefit descriptions** varied widely in phrasing, requiring significant preprocessing and standardization efforts to ensure analytical accuracy.
- Some **statistical tests (e.g., t-tests, chi-squared)** assume normality or independence, which may not fully apply to real-world job market data distributions.
- **External cost of living (COL) indices** could not be integrated, as our dataset already included pre-normalized salary fields, limiting regional adjustment flexibility.
- A few dataset columns had **missing, vague, or malformed location values**, making geographic comparisons occasionally noisy or incomplete.

Conclusion

The LinkedIn job postings dataset gave us a rich view of today's labor market. We explored how salary is influenced by multiple factors like experience level, job title, skill type, and even benefits such as 401(k) or medical insurance. By using machine learning models like XGBoost and clustering techniques on job descriptions, we identified patterns and trends that go beyond job titles—uncovering hybrid roles, skill-driven pay gaps, and salary prediction drivers.

We also analyzed broader trends such as the rise of remote work, salary transparency across states, cost-of-living-adjusted compensation, and how seasonality impacts hiring. Text analytics helped us discover the power of language—feminine-coded and empathetic terms increased applicant interest, while vague phrases like "competitive salary" reduced engagement. These subtle language choices proved to be powerful levers in improving equity and application rates.

Overall, our findings can guide job seekers toward high-value roles, help companies write more inclusive and attractive job listings, and support policymakers focused on workplace fairness. This project shows how data, when analyzed through multiple lenses, can shape a deeper understanding of the evolving job market and provide practical tools for smarter workforce decisions.

Explore More on Our Blog:

For a deeper dive into our findings and reflections, check out our blog on Medium:

https://medium.com/@anwar.r.752/e2f2f2b85231

We've translated complex data into meaningful insights — presented with clarity, curiosity, and creativity. It's not just about numbers; it's about the story they tell.